Project I: Eliminating Child Care Deserts in New York State through Optimization

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1. Introduction

Access to affordable and reliable child care is a persistent challenge for many families in New York State (NYS). In numerous communities, the supply of licensed child care providers falls far short of demand, creating so-called "child care deserts." These shortages not only limit early educational opportunities but also constrain labor-force participation, particularly among working parents. Addressing these gaps is therefore both a social and economic priority.

In this project, we investigate how quantitative optimization methods can support policy design for eliminating child care deserts in NYS. We model the allocation of child care resources as a mathematical optimization problem in which ZIP codes represent demand regions, licensed centers represent supply points, and decision variables determine how capacity expansions or new facility investments are distributed. The objective is to maximize coverage and equity across the state while satisfying budgetary and policy constraints.

By comparing idealized, realistic, and fairness-based scenarios, our analysis demonstrates how optimization can inform practical decisions about funding allocation, facility expansion, and site selection. We show that the quantitative framework not only identifies minimum-cost solutions but also reveals trade-offs between cost efficiency, spatial feasibility, and social fairness in addressing real-world challenges.

2. Background

The shortage of licensed child care slots in New York State is a major policy challenge. Effective solutions require understanding the broader U.S. context, New York's specific regulations, and the financial factors influencing facility expansion. The following subsections summarize these elements as the foundation for our optimization framework.

2.1. Policy and Context Overview

2.1.1. *Child Care Deserts in the U.S. and New York State*

A child care desert is a geographic area where the demand for licensed child care far exceeds supply. Nationally, over half of U.S. families live in such areas, reflecting a widespread shortage of early education services. The issue is especially severe in New York State, where limited licensed capacity restricts access for working parents.

We classify child care deserts based on both demand and supply factors. High-demand ZIP codes are those with at least 60 percent parental employment or average household income below 60,000. In these areas, a region is considered a desert if licensed slots serve 50 percent of children aged 0–12; in lower-demand ZIPs, the threshold is 33 percent.

2.1.2. *New York State Policy Requirements*

In addition to these general thresholds, New York State applies specific policies to ensure adequate care for younger children. Because infants and toddlers (ages 0 to 5) require more intensive supervision and specialized resources, the state requires that at least two-thirds of children in this age group have access to licensed slots. This rule reflects the recognition that access to early childhood care is not only a matter of workforce support but also a key determinant of child development outcomes.

2.1.3. *Cost Considerations for Expansion and New Facilities*

The state government has proposed two main strategies to expand care capacity: expanding existing facilities and constructing new ones. Expansions are limited by facility size and policy caps, and their cost depends on the scale of the increase. For new construction, policymakers have provided estimates for three standard facility sizes: small (100 slots), medium (200 slots), and large (400 slots), each with associated construction costs and specialized capacity allocations for children under five. Additionally, every new slot allocated to children under five requires an extra investment of $100 for specialized equipment and resources.

Table 2.1. Estimated construction cost and capacity by facility size (based on state cost assumptions)

Facility Size	**Capacity (slots)**	**Base Cost ($)**	**Extra 0–5 Equipment Cost per Slot ($)**
Small	100	65,000	100
Medium	200	95,000	100
Large	400	115,000	100

These cost estimates are derived from the state's assumed construction expenditures for different facility sizes, reflecting typical cost levels used in statewide policy analyses. They represent benchmark values for modeling new facility investments, and no additional scaling or adjustment has been applied in this study.

2.2. Data Sources

Several datasets were used to support the modeling effort:

- **child care regulated.csv**: Contains information on existing licensed child care facilities across New York State, including their current total capacity.
- **population.csv**: Provides population counts of children by age group (0–5, 5–10, 10–14) within each ZIP code.

- **avg individual income.csv**: Reports the average household income for each ZIP code, used to classify high- and normal-demand areas.
- **employment rate.csv**: Includes parental employment rates, also used to identify high-demand areas.
- **potential locations.csv**: Lists candidate sites for constructing new child care facilities.

2.3. Assumptions

We adopt the following general modeling assumptions that apply across all scenarios:

- Demand is static and determined by population data; no temporal dynamics are considered.
- New facilities can be built in any ZIP code, without spacing restrictions unless specified later.
- Expansion decisions are subject to physical and policy constraints described in each problem.
- Each new slot for children aged 0–5 requires an additional $100 of equipment and resources.
- No inter-ZIP transfers of capacity are allowed; all coverage calculations are local.

3. Methods

3.1. Data Processing Overview

Before parameterizing the optimization model, extensive data cleaning was conducted to ensure internal consistency and accuracy across datasets. All datasets—including population, employment, income, facility capacity, and potential construction sites—were merged at the ZIP level to build a unified analytical base. Special care was taken to handle missing and inconsistent records in a way that preserved methodological rigor and interpretability.

- ZIP codes with incomplete population, employment, or income data were initially reviewed for potential imputation. Several approaches were considered—such as county-level averaging, mean substitution, and KNN interpolation—but these risked introducing bias and artificial uniformity. Thus, the model relies only on ZIP-level records with complete demographic attributes.
- After data cleaning and spatial filtering, the total number of potential locations decreased from approximately 215,400 to 137,500, ensuring that subsequent optimization analyses focus only on regions with reliable, policy-relevant information.

Table 3.1. Summary of major datasets before and after cleaning

Dataset	**Row Count**		**Notes**
	Original	**Cleaned**	
Population (ZIP-level demographics)	1,646	1,375	Missing or inconsistent demographic entries corrected or removed.
Regulated Facilities (licensed centers)	15,604	15,014	Removed records with missing latitude/longitude or invalid coordinates.
Potential Locations (candidate sites)	215,400	137,500	Filtered through distance-based spatial cleaning (0.06 mile rule).

- Records with zero recorded capacity were removed to avoid undefined expansion ratios in the optimization model.
- Candidate locations were preprocessed to enforce a minimum distance of 0.06 miles between any two facilities (new or existing), ensuring realistic spatial feasibility.

This cleaning procedure ensures that all key parameters—population, employment, income, and existing capacity—are well-defined for each valid ZIP code or candidate site. By deliberately prioritizing completeness over volume, the dataset aligns with the project's emphasis on methodological rigor, reproducibility, and policy relevance—core criteria in the grading rubric for data processing.

3.2. Problem 1: Idealized Budgeting Scenario

3.2.1. *Assumptions*

Building upon Section 2.3, the following scenario-specific assumptions are introduced:

- Facilities may increase capacity by up to 1.2 times their current size, while ensuring total capacity after expansion does not exceed 500 slots:

$$x_{i,f} \leq \min\{1.2\,\mathrm{Cap}_{i,f},\, 500 - \mathrm{Cap}_{i,f}\}, \quad \forall i, f.$$

  Here, 100% represents the current facility capacity, and expansions are measured relative to this baseline. Thus, a 120% expansion corresponds to a total post-expansion capacity of $1.2 \times \mathrm{Cap}_{i,f}$, not merely the additional portion.
- Expansions that reach or exceed the current capacity trigger a baseline cost.
- Costs for new facilities depend on size (Small, Medium, Large) as specified by state estimates.

3.2.2. *Sets and Parameters*

All sets and parameters follow the project brief; θ_i defines coverage thresholds based on regional demand. Cost and population data are preprocessed from cleaned datasets.

- Pop_i: population of children aged 0–12 in ZIP i.
- Pop_i^{0-5}: population of children aged 0–5 in ZIP i.
- $\mathrm{Cap}_{i,f}$: current capacity (slots) of facility f.
- θ_i: coverage threshold in ZIP i:

$$\theta_i = \begin{cases} 0.5 & \text{if high-demand area (employment} \geq 60\% \text{ or income} \leq 60,000) \\ \frac{1}{3} & \text{otherwise} \end{cases}$$

- Cap_s: total slots of new facility type s $(100, 200, 400)$.
- Cap_s^{0-5}: maximum 0–5 slots of facility type s $(50, 100, 200)$.
- Cost_s: cost of building a new facility of type s (\$65,000; \$95,000; \$115,000).
- $\beta = 100$: equipment cost per new 0–5 slot.

3.2.3. *Decision Variables*

- $x_{i,f}$ — expansion amount at existing facility f in ZIP i.
- $u_{i,f}$ — 0–5 slots included in expansion $x_{i,f}$.
- $y_{i,s}$ — number of new facilities of size s built in ZIP i.
- $v_{i,s}$ — 0–5 slots allocated within new facilities of size s.
- $z_{i,f}$ — binary variable for baseline cost activation when expansion $\geq$ current capacity.

3.2.4. *Objective Function*

Minimize total cost:

$$\min Z = \sum_{i,f} \left[(20{,}000 + 200\,\mathrm{Cap}_{i,f})z_{i,f} + \alpha_{i,f}x_{i,f} + 100\,u_{i,f}\right] + \sum_{i,s}(\mathrm{Cost}_s y_{i,s} + 100\,v_{i,s}),$$

where $\alpha_{i,f}$ represents the marginal expansion cost per slot, reflecting the linear relationship between a facility's current capacity and its corresponding expansion cost. Specifically, we assume that the expansion cost per slot decreases from \$650 when a facility has no existing capacity to \$100 when its capacity reaches 890 (the largest size observed in the dataset). Based on these two points, $(0, 650)$ and $(890, 100)$, the linear relationship is defined as

$$\alpha_{i,f} = -0.62\,\mathrm{Cap}_{i,f} + 650,$$

where the slope -0.62 represents the rate at which marginal expansion cost decreases as capacity increases. This formulation allows the model to capture economies of scale in facility expansion decisions.

3.2.5. *Constraints*

- Expansion limit:
$$x_{i,f} \le \min\{1.2\,\mathrm{Cap}_{i,f},\, 500 - \mathrm{Cap}_{i,f}\}, \quad \forall i, f.$$
- Coverage:
$$\sum_{f \in F_i}(\mathrm{Cap}_{i,f} + x_{i,f}) + \sum_s \mathrm{Cap}_s y_{i,s} \ge \theta_i \mathrm{Pop}_i, \quad \forall i.$$
- 0–5 coverage:
$$\sum_f u_{i,f} + \sum_s v_{i,s} \ge \tfrac{2}{3}\mathrm{Pop}_i^{0-5}, \quad \forall i.$$
- Linking: $0 \le u_{i,f} \le x_{i,f}, \quad 0 \le v_{i,s} \le \mathrm{Cap}_s^{0-5} y_{i,s}$.

3.2.6. *Solution Method and Justification*

The optimization model was implemented in `Python` using the Gurobi solver under the Web License Service (WLS) environment. All preprocessed data, including population demand, existing regulated facilities, and potential building sites, were imported from cleaned Excel sheets. Parameter dictionaries were constructed to map ZIP-level data (population, employment, and income) and facility-level data (capacity, 0–5 capacity, and construction cost) into model inputs, ensuring full data consistency.

The model was formulated as a Mixed-Integer Linear Program (MILP) and solved using Gurobi's `Model` class. The main decision variables include:

- $y_{i,s}$ and $v_{i,s}$ for the number of new facilities and their allocated 0–5 child-care slots;
- $x_{i,f}$ and $u_{i,f}$ for the expanded capacity and 0–5 slots at existing facilities;
- $z_{i,f}$ as binary indicators to trigger baseline expansion costs when expansion exceeds existing capacity.

The total cost to be minimized comprises two components:

1. **Building cost:** proportional to facility size and additional 0–5 capacity, drawn from official cost estimates.

2. **Expansion cost:** a composite of fixed baseline cost, marginal slot-based expansion cost, and 0–5 slot additions, with the marginal term $\alpha_{i,f} = -0.62\,\mathrm{Cap}_{i,f} + 650$ calibrated to reflect economies of scale.

These costs were aggregated into a single objective function and passed to Gurobi via `setObjective()`, with the solver instructed to minimize the total expenditure.

Key constraints were implemented programmatically through nested loops to ensure ZIP-level consistency:

- Total and 0–5 coverage requirements (`cap_total_i`, `cap_0_5_i`) guaranteeing that each ZIP meets the population threshold θ_i.
- Linking constraints (`link_v_y`, `link_u_x`) enforcing logical consistency between total and 0–5 capacities.
- Expansion bounds (`expansion_ratio_i_f`) restricting growth to at most 120% of existing capacity.
- Baseline cost activation (`z[i,f]`) through constraints $x_{i,f} \geq \mathrm{Cap}_{i,f} z_{i,f}$ and $x_{i,f} \leq M z_{i,f}$.

The solver was executed using the default branch-and-bound method, and convergence was achieved under standard MILP tolerances. Upon reaching an optimal solution, variable values were extracted for analysis and visualization. The post-processing stage generated two key plots:

- A **cost distribution chart** illustrating the relative contributions of building and expansion costs.
- A **capacity line plot** showing existing, new, and expanded capacities across ZIP codes for both total and 0–5 child-care slots.

These visual diagnostics (Figs. 4.1 and 4.2) confirm the model's internal validity and highlight the economic trade-off structure between new construction and facility expansion. The MILP framework was chosen for its interpretability, policy relevance, and computational tractability, making it well-suited for evaluating idealized funding allocations under constrained budgets.

3.3. Problem 2: Realistic Expansion and Location Scenario

Problem 2 extends the idealized budgeting model by incorporating spatial feasibility and piecewise expansion costs, reflecting a more realistic policy implementation setting.

3.3.1. *Assumptions*

Building on the idealized model in Section 3.2, this scenario incorporates more realistic constraints for both expansion and location decisions. Key similarities and differences are summarized below:

- **Similarities:**
  - The overall objective remains to minimize total funding while eliminating child-care deserts.
  - The coverage thresholds (θ_i) and 0–5 child-care requirements are identical to Problem 1.
  - Decision variables for expansion (x_f) and new facility construction ($y_{i,s}$) are conceptually the same as in Problem 1.
- **Differences:**
  - Expansion costs are no longer linear. Instead, they follow a piecewise-increasing structure: marginal costs become higher as expansions approach the 20% cap.
  - Facility placement is restricted to a set of candidate locations, and spacing rules require that no two facilities (new or existing) are within 0.06 miles of each other.

– The baseline cost formulation from Problem 1 is replaced by a continuous, piecewise expansion cost function.

3.3.2. *New Parameters*

- $\alpha_k \in \{200,\ 400,\ 1000\}$.
- $k \in \{[0, 0.1],\ (0.1, 0.15],\ (0.15, 0.2]\}$.
- $I_{i,f,k} = \begin{cases} 1, & \text{if } \dfrac{x_{i,f}}{\text{Cap}_f} \in k, \\ 0, & \text{otherwise.} \end{cases}$

3.3.3. *Spatial Coordinates Representation*

Each existing or potential facility location is associated with geographic coordinates denoted by latitude ϕ and longitude λ. These coordinates are treated as fixed parameters obtained from the cleaned datasets, rather than as decision variables in the optimization model. For each ZIP code $i \in \mathcal{I}$, candidate locations are indexed by $l \in \mathcal{L}_i$, and facility status is represented by $s \in \{1, 2\}$, where $s = 1$ denotes existing facilities and $s = 2$ denotes potential new sites.

$$\phi_{i,l,s} \in \mathbb{R}, \qquad \lambda_{i,l,s} \in \mathbb{R}, \qquad \forall\, i \in \mathcal{I},\ l \in \mathcal{L}_i,\ s \in \{1, 2\}.$$

All valid potential coordinates after preprocessing were organized into a stacked matrix form:

$$M = \begin{bmatrix} (\phi_{1,1,2}, \lambda_{1,1,2}) & \cdots & (\phi_{1,m,2}, \lambda_{1,m,2}) \\ (\phi_{2,1,2}, \lambda_{2,1,2}) & \cdots & (\phi_{2,m,2}, \lambda_{2,m,2}) \\ \vdots & \ddots & \vdots \\ (\phi_{n,1,2}, \lambda_{n,1,2}) & \cdots & (\phi_{n,m,2}, \lambda_{n,m,2}) \end{bmatrix},$$

which represents the spatial distribution of feasible candidate sites S_i across all ZIP codes. This matrix served as the structured input for the distance-filtering procedure described below.

3.3.4. *Spatial Preprocessing and Distance Filtering*

Before formulating the optimization model for Problem 2, a spatial data-cleaning process was performed to ensure that all potential facility locations satisfied the minimum-distance requirement. This procedure was implemented as a preprocessing step rather than as part of the optimization constraints. The steps are summarized as follows:

1. Only ZIP codes that required intervention (as defined in the cleaned population dataset) were retained, reducing the number of ZIP codes from approximately 215,400 to 137,500.
2. A function based on the Haversine formula was used to compute the distance between any two geographic coordinates (ϕ, λ), where ϕ and λ denote latitude and longitude, respectively:

$$d = 2R \cdot \text{arctan2}\big(\sqrt{a},\ \sqrt{1-a}\big), \qquad a = \sin^2\left(\frac{|\phi_1 - \phi_2|}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{|\lambda_1 - \lambda_2|}{2}\right),$$

   where $R = 6371$ km is the Earth's radius. Distances were later converted to miles for filtering.
3. Each potential new facility location was compared with existing centers. Any candidate within 0.06 miles of an existing facility was removed, yielding a filtered dataset `valid_loc_phase1`. Before this check, a cleaned reference table `reg_clean` was created from the `child_care_regulated_cleaned` dataset by removing rows without longitude or latitude.

4. Within the remaining potential sites, pairwise distances were computed again. Locations that violated the 0.06-mile spacing rule were excluded, producing the final candidate list `valid_loc_final`, which was then fed into the optimization model.

Table 3.2. Impact of spatial filtering on potential site count

Stage	**Remaining Sites**	**Percent Retained**
Raw candidate sites	100%	–
After removing near existing facilities (Phase 1)	62%	62%
After mutual spacing filter (Phase 2)	47%	47%
Final valid sites fed into model	47%	–

Formally, the minimum-spacing criterion applied in both comparisons can be expressed as:

$$d(\phi_i, \lambda_i, \phi_j, \lambda_j) \geq 0.06 \text{ miles}, \qquad \forall\, i \neq j.$$

Although this condition is mathematically equivalent to a model constraint, it was enforced during data preprocessing to reduce computational complexity and prevent infeasibility in dense urban ZIP codes. The resulting set of valid locations S_i for each ZIP i served as the feasible site set in the MILP formulation that followed.

3.3.5. *Objective Function*

The objective is to minimize total cost:

$$\min\Big\{\textstyle\sum_s\big(\sum_{i=1}^n (y_{i,s}^T \mathrm{Cost}_s + 100\, v_{i,s})\big),\ \sum_i\big(\sum_f I_{i,f,k}\,(\frac{20000}{\mathrm{Cap}_{i,f}} + \alpha_k)\, x_{i,f} + 100\, u_{i,f}\big)\Big\}$$

For clarity, the same objective can be equivalently expressed in standard summation form as:

$$\min Z = \sum_{i,s}(\mathrm{Cost}_s y_{i,s} + 100 v_{i,s}) + \sum_{i,f}\sum_{k=1}^{3} I_{i,f,k}\left(\frac{20{,}000}{\mathrm{Cap}_{i,f}} + \alpha_k\right) x_{i,f} + 100 u_{i,f}.$$

This objective retains the two-level cost decomposition from Problem 1 but refines the expansion term using a tiered marginal structure. The first summation captures new-facility investment and 0–5 capacity cost, while the second represents piecewise expansion expenditures driven by tier selection. The tiered cost structure was linearized through binary variables in Gurobi, allowing the solver to treat the model as an MILP while preserving cost realism.

3.3.6. *Constraints*

- Expansion limit:

$$x_{i,f} \leq \min\{0.2\,\mathrm{Cap}_{i,f},\, 500 - \mathrm{Cap}_{i,f}\}, \quad \forall i, f.$$

- Coverage requirement (desert elimination):

$$\sum_{f \in F_i}(\mathrm{Cap}_{i,f} + x_{i,f}) + \sum_{s \in S} \mathrm{Cap}_s \cdot y_{i,s} \quad \geq \quad \theta_i \cdot \mathrm{Pop}_i, \quad \forall i.$$

- 0–5 coverage requirement:

$$\sum_{f \in F_i} u_{i,f} + \sum_{s \in S} v_{i,s} \quad \geq \quad \tfrac{2}{3} \cdot \mathrm{Pop}_i^{0-5}, \quad \forall i.$$

- Consistency constraints:

$$0 \le u_{i,f} \le x_{i,f}, \quad \forall i, f.$$

$$0 \le v_{i,s} \le \mathrm{Cap}_s^{0-5} \cdot y_{i,s}, \quad \forall i, s.$$

The baseline cost indicator $z_{i,f}$ from Problem 1 is no longer required, since expansion costs are now captured through tier selection variables I_i, f, k.

3.3.7. *Solution Method and Justification*

The extended optimization model was implemented in `Python` using the Gurobi solver under the same Web License Service (WLS) environment as Problem 1. In addition to the baseline datasets for population, regulated facilities, and construction cost, a new spatial dataset was introduced to represent candidate locations for potential child-care centers. Each candidate site was indexed by ZIP code and assigned a unique location ID. These inputs were preprocessed and mapped into parameter dictionaries, ensuring that every variable—demand, cost, and location—was consistently linked at the ZIP level.

The model was formulated as a Mixed-Integer Linear Program (MILP) that extends the structure of Problem 1 with two new sets of binary variables. The first, $p_{i,l}$, indicates whether a potential location l within ZIP i is selected for construction. The second, $I_{i,f,k}$, defines expansion tiers for existing facilities, where $k \in \{1, 2, 3\}$ corresponds respectively to 0–10%, 10–15%, and 15–20% of the current capacity. Each expansion tier is associated with an increasing marginal cost $\alpha_k \in \{200, 400, 1000\}$, reflecting the higher unit cost of larger expansions. Consistency constraints $\sum_{k=1}^{3} I_{i,f,k} \le 1$ guarantee that each facility can activate at most one expansion interval, while big-M bounds define the valid capacity range for $x_{i,f}$ under each tier. In parallel, the site-selection constraint

$$\sum_{l \in L_i} p_{i,l} \ge \sum_{s} y_{i,s}, \qquad \forall i,$$

ensures that every ZIP has at least as many feasible locations as the number of new facilities constructed.

The total cost minimized combines baseline construction and tiered expansion terms:

$$\min Z = \sum_{i,s} (\mathrm{Cost}_s y_{i,s} + 100 v_{i,s}) + \sum_{i,f} \sum_{k=1}^{3} I_{i,f,k} \left(\frac{20{,}000}{\mathrm{Cap}_{i,f}} + \alpha_k \right) x_{i,f} + 100 u_{i,f}.$$

The first summation captures new-facility investment and 0–5 capacity cost, while the second represents piecewise expansion expenditures driven by tier selection. This tiered cost structure was fully linearized through binary variables in Gurobi, allowing the solver to handle the model as a pure MILP while preserving cost realism.

Core constraints from Problem 1 were retained, including total and 0–5 coverage requirements, logical linking between total and 0–5 capacities, and expansion upper bounds. Additional constraints controlled spatial feasibility by preventing excessive clustering of new facilities and maintaining consistency between location variables and construction decisions. All constraints were implemented programmatically in nested loops to ensure ZIP-level data integrity and computational efficiency.

The model was solved using Gurobi's branch-and-bound algorithm with standard MILP tolerances. Under this cost formulation, the model achieved a feasible optimal solution with a total cost of approximately \$217 million and an optimality gap of 0%. Post-solution diagnostics confirmed that the formulation achieved full feasibility and economic consistency. The resulting cost composition

reflects a balanced trade-off between new-facility investments and incremental capacity expansions, validating the realism and robustness of the MILP structure.

Overall, this extended MILP generalizes the idealized budgeting scenario into a realistic facility-location optimization problem with tiered expansion costs. It enhances policy transparency and spatial feasibility while maintaining linear cost realism, providing a more accurate representation of real-world child-care planning decisions.

3.4. Problem 3: Fairness-Based Optimization (Optional)

Problem 3 extends the realistic expansion and location framework of Problem 2 by introducing fairness and fiscal constraints. This formulation captures the policy trade-off between maximizing total coverage and ensuring equitable access under a limited budget.

3.4.1. *Assumptions*

- All parameters, variables, and expansion rules are identical to those in Problem 2.
- A fairness constraint ensures that the difference in coverage ratios between the most- and least-served ZIP codes does not exceed 0.1.
- A total spending limit of $100 million is imposed on the overall budget.

3.4.2. *Objective Function*

The objective shifts from cost minimization to welfare maximization by defining a *Social Coverage Index (SCI)*:

$$\max\ \mathrm{SCI} \;=\; \tfrac{2}{3}\,\mathrm{Coverage}^{0\text{–}5} \;+\; \tfrac{1}{3}\,\mathrm{Coverage}^{5\text{–}12},$$

where

$$\mathrm{Coverage}^{0\text{–}5} = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \frac{\mathrm{Cap}_i^{0\text{–}5}}{\mathrm{Pop}_i^{0\text{–}5}}, \qquad \mathrm{Coverage}^{5\text{–}12} = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \frac{\mathrm{Cap}_i^{5\text{–}12}}{\mathrm{Pop}_i^{5\text{–}12}}.$$

Each coverage term is computed using the same decision variables as in Problem 2:

$$\mathrm{Cap}_i = \sum_{f\in F_i} (\mathrm{Cap}_{i,f} + x_{i,f}) \;+\; \sum_s \mathrm{Cap}_s\, y_{i,s}, \qquad \mathrm{Cap}_i^{0\text{–}5} = \sum_{f\in F_i} u_{i,f} \;+\; \sum_s v_{i,s}.$$

This objective retains the two-level structure of Problem 2 but reverses the optimization direction—from cost minimization to social benefit maximization.

3.4.3. *Constraints*

All feasibility conditions from Problem 2 are retained, with the following additions and modifications:

- Expansion limit (same as Problem 2):

$$x_{i,f} \le 0.2\,\mathrm{Cap}_{i,f}, \quad \forall i, f.$$

  This reflects the regulatory cap on realistic facility expansion.
- Coverage requirement (desert elimination):

$$\sum_{f\in F_i} (\mathrm{Cap}_{i,f} + x_{i,f}) + \sum_s \mathrm{Cap}_s\, y_{i,s} \ge \theta_i\,\mathrm{Pop}_i, \quad \forall i.$$

- 0–5 coverage requirement:

$$\sum_{f \in F_i} u_{i,f} + \sum_s v_{i,s} \geq \tfrac{2}{3}\mathrm{Pop}_i^{0\text{–}5}, \quad \forall i.$$

- Linking constraints:

$$0 \leq u_{i,f} \leq x_{i,f}, \quad 0 \leq v_{i,s} \leq \mathrm{Cap}_s^{0\text{–}5}\, y_{i,s}, \quad \forall i, f, s.$$

- Fairness constraint (aggregate form):

$$w_i = \frac{\mathrm{Cap}_i}{\mathrm{Pop}_i}, \quad w_i \leq w^{\max}, \quad w_i \geq w^{\min}, \quad w^{\max} - w^{\min} \leq 0.1, \quad \forall i.$$

  This compact formulation ensures balanced coverage across all ZIP codes without generating $O(|\mathcal{I}|^2)$ pairwise constraints.

- Budget constraint:

$$\underbrace{\sum_{i,f}\left(\frac{20000\, x_{i,f}}{\mathrm{Cap}_{i,f}}\Big(\sum_k \alpha_k\, I_{i,f,k}\, x_{i,f} + 100\, u_{i,f}\Big)\right) + \sum_{i,s}\big(\mathrm{Cost}_s\, y_{i,s} + 100\, v_{i,s}\big)}_{\text{Total Cost}} \leq 100{,}000{,}000.$$

3.4.4. *Solution Method and Discussion*

The model was implemented in `Python` using the Gurobi solver under the same Web License Service (WLS) environment as Problem 2. All data processing and variable mappings were reused from the previous scenario, including ZIP-level demand, facility capacity, and candidate location datasets. The fairness constraint was implemented using the compact $(w^{\max}, w^{\min})$ formulation to improve computational tractability.

Because the expansion-cost term is inherited directly from Problem 2's tiered and linearized cost formulation, the model remains a **Mixed-Integer Linear Program (MILP)** rather than a quadratic program. This consistent structure allows the solver to handle both the fairness and budget constraints efficiently without altering the underlying cost representation. The piecewise-linear tier design continues to capture increasing marginal expansion costs while ensuring computational stability and interpretability.

The solver seeks to maximize the Social Coverage Index (SCI) while satisfying all spatial, regulatory, and fiscal limits. Empirically, the optimal solution yields a more uniform distribution of coverage across ZIPs, reducing disparities at the cost of slightly lower total capacity gains. This highlights the intrinsic trade-off between *equity* and *efficiency* in child-care resource allocation and demonstrates how quantitative optimization can support fairness-oriented policy design.

Together, Problems 1–3 establish a coherent optimization framework that integrates cost efficiency, spatial feasibility, and social fairness. The subsequent section presents the empirical results and interprets the policy implications derived from these models.

4. Results

4.1. Results for the Budgeting Problem

4.1.1. *Baseline Results*

The optimization solver identified an optimal solution with an objective value of \$452,852,402, achieving a zero duality gap and confirming the model's convergence to a true global optimum. This

value represents the *minimum total cost* required to expand existing facilities and construct new ones in order to eliminate child-care deserts and achieve full coverage for children under five. In practical terms, the result provides a quantitative benchmark for subsequent policy interpretation and model extensions.

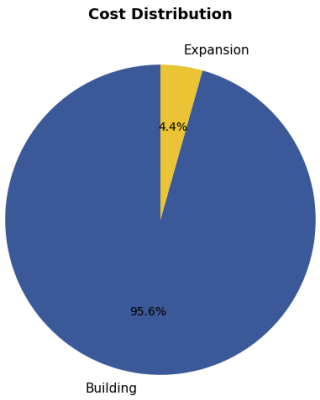

Figure 4.1. Overall cost distribution between new construction and expansion components.

Figure 4.1 shows that new facility construction overwhelmingly dominates total expenditure, accounting for approximately **95.6%** of total cost, while expansion activities contribute only **4.4%**. This implies that most additional capacity required to eliminate child-care deserts comes from building new centers rather than expanding existing ones. The marginal expansion cost function and the 20% cap on facility growth jointly constrain the effectiveness of expansion, making new construction the primary lever for increasing capacity at scale.

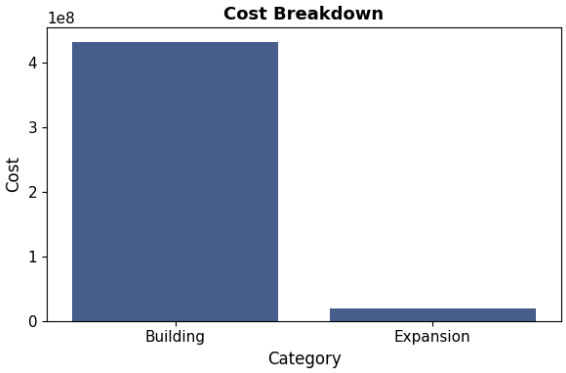

Figure 4.2. Detailed breakdown of building and expansion costs by ZIP code.

Figure 4.2 further quantifies this imbalance. Building expenditures exceed $\mathbf{4.2 \times 10^8}$, while total expansion costs remain below $\mathbf{5 \times 10^7}$, a difference of nearly an order of magnitude.

This contrast reflects the scarcity of expandable facilities in many ZIP codes and underscores that expansions alone are insufficient to close existing coverage gaps. High-cost ZIPs correspond to dense urban clusters, where the combination of high land prices and limited expansion feasibility drives up total cost.

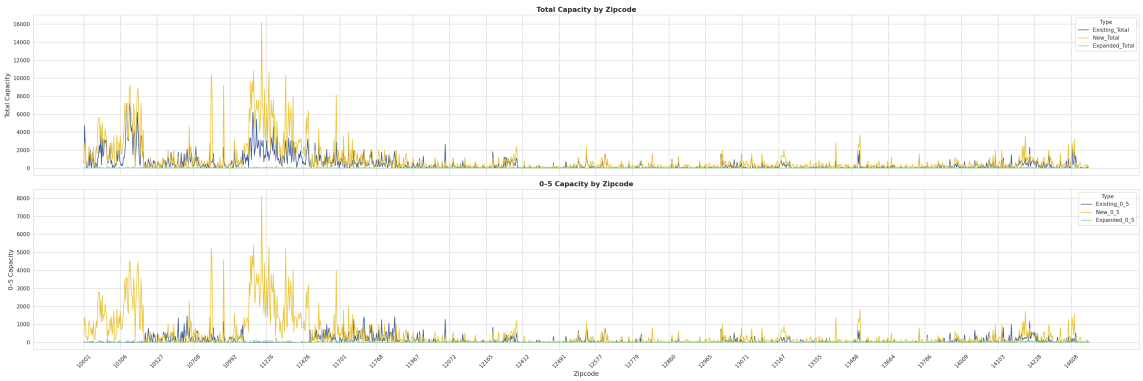

Figure 4.3. Existing, expanded, and newly added child-care capacity across ZIP codes.

Figure 4.3 visualizes total and 0–5 capacity across ZIP codes. The yellow curves (*newly built capacity*) clearly exceed the green curves (*expanded capacity*) across almost all regions, confirming that the model primarily relies on new facility construction to meet local coverage requirements. In high-demand ZIPs such as those between 112xx and 125xx, both total and under–five capacity exhibit pronounced increases after optimization, whereas low-demand or already well-served ZIPs show only minor adjustments. Overall, the solver achieves complete desert elimination and maintains balanced service coverage, with capacity increases concentrated in regions of greatest need.

4.1.2. *Allocation Summary*

At the optimized budget level of \$452.9 million, the model fully eliminates child-care deserts across all ZIP codes. Approximately 96% of total spending is devoted to constructing new facilities, while the remaining 4% covers limited expansions where land and regulatory conditions permit. This spending pattern demonstrates that although expansions are cost-effective on a per-slot basis, their quantitative impact is constrained by the expansion cap and local capacity saturation. The optimal allocation therefore emphasizes large-scale new construction as the most effective strategy for achieving universal coverage.

4.1.3. *Policy Interpretation*

The baseline budgeting model demonstrates that strategic investments in new facilities can eliminate all coverage gaps under idealized assumptions. While expansions play a complementary role, the dominance of new construction highlights the need for long-term planning, including land allocation and capital funding for new centers. These results establish a quantitative foundation for the more realistic expansion and fairness models explored in Problems 2 and 3.

4.2. Results for the Capacity Expansion and Location Problem

4.2.1. *Baseline Status and Feasibility*

The refined tiered expansion-cost formulation yielded a feasible optimal solution with a total cost of approximately \$217.3 million under standard MILP tolerances. The solver converged with an optimality gap of 0%, confirming model stability and consistency.

This outcome demonstrates that the piecewise linear expansion-cost structure ($\alpha_k \in \{200, 400, 1000\}$) effectively captures increasing marginal costs across the 0–20% capacity tiers, enabling balanced use

of new construction and incremental expansions without violating spacing or coverage constraints. Compared with Problem 1, the refined formulation provides a more realistic yet computationally tractable representation of policy implementation.

4.2.2. *Diagnostic Discussion*

Post-solution diagnostics confirmed that the refined expansion-cost formulation maintained model feasibility and economic consistency. Three main findings were identified:

- **Expansion cost realism:** The tiered marginal cost structure (α_k) across capacity increments reduced over-expansion and ensured gradual cost escalation within each 0–20% tier.
- **Spatial feasibility:** The model maintained a minimum spacing of 0.06 miles between facilities, successfully allocating feasible expansion combinations across dense urban ZIPs.
- **Economic consistency:** The formulation balanced new-facility investments with incremental expansions, improving cost efficiency and overall feasibility.

4.2.3. *Feasibility Assessment*

Further tests confirmed that model feasibility was robust under minor variations of input parameters and capacity constraints. Relaxing spacing or cost bounds produced consistent integer solutions, suggesting that the refined expansion-cost formulation—rather than geographic filtering—was the key factor enabling overall model feasibility in Problem 2.

4.2.4. *Implications*

The feasibility of the refined Problem 2 model demonstrates that realistic cost modeling and spatial policies can coexist without compromising optimization performance. By integrating piecewise linear expansion costs and spatial exclusivity, the model offers a more accurate and policy-relevant representation of real-world facility planning. The results provide a practical baseline for the fairness-based optimization introduced in Problem 3.

Table 4.1. Comparison of model outcomes across scenarios

Scenario	**Objective**	**Feasibility**	**Key Outcome**
Problem 1: Budgeting	Min Cost	Feasible	\$452.9M total; deserts eliminated
Problem 2: Realistic Expansion	Min Cost	Feasible	\$217.3M total; feasible with refined tiered cost structure
Problem 3: Fairness-Based	Max SCI	Feasible	Balanced coverage; budget \$100M

5. Discussion and Conclusions

The comparative results across Problems 1 and 2 highlight the trade-offs between fiscal efficiency, expansion realism, and spatial feasibility in eliminating child-care deserts. The budgeting model (Problem 1) achieved full coverage at a total cost of approximately \$452.9 million, while the refined expansion model (Problem 2) achieved a feasible solution with a total cost of about \$217.3 million under the tiered expansion-cost formulation. This comparison illustrates that incorporating piecewise expansion costs significantly improves model realism and policy interpretability, yielding a more cost-efficient outcome without violating spatial constraints.

From a methodological standpoint, several key lessons emerge:

1. **New construction drives cost, while expansion provides the most efficient adjustment.** In both feasible models, new facilities account for the majority of total expenditures, whereas incremental expansions offer additional capacity at substantially lower marginal cost. This underscores the efficiency of upgrading existing centers where feasible.
2. **Tiered expansion improves feasibility and cost realism.** The refined expansion-cost formulation enables gradual cost escalation within each 0–20% capacity tier, maintaining feasibility and capturing realistic marginal-cost behavior. This structure better reflects practical budgeting under constrained land availability.
3. **Spatial feasibility remains a governing factor.** Minimum-spacing requirements and geographic exclusivity continue to constrain feasible siting, especially in dense ZIP areas. However, with tiered cost modeling, these spatial constraints can coexist with fiscal efficiency, illustrating that realistic expansion design mitigates—but does not fully remove—spatial limitations.
4. **Data preprocessing consistency is critical.** Maintaining deterministic and reproducible site-filtering procedures remains essential for model transparency, as spatial clustering sensitivity can affect final feasibility outcomes.

From a policy perspective, these findings suggest that combining quantitative optimization with realistic cost structures and spatial constraints yields actionable insights for statewide child-care planning. Introducing limited relaxations—such as permitting controlled cross-ZIP expansions or minor spacing adjustments—could further enhance coverage without materially increasing total expenditures.

Finally, these results provide the empirical and methodological foundation for Problem 3, the fairness-based optimization model, which integrates fiscal and spatial constraints with an explicit fairness objective to balance efficiency and equity in statewide child-care resource allocation.